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Experiment No. 03

Aim: To deploy a multi-service application using Docker Compose.

Theory:

Docker Compose is a tool provided by Docker that is used to define and run multi-container applications using a single YAML configuration file called docker-compose.yml. Instead of creating and managing each container individually using separate docker run commands, Docker Compose allows multiple services to be configured and started together with a single command.

In modern applications, multiple services such as a web server, a database server, and administration tools need to work together. Docker Compose simplifies this process by defining all services, images, ports, environment variables, volumes, and dependencies in one file. It automatically creates a common network so that services can communicate using their service names instead of IP addresses.

In this experiment, three services are deployed:

- **Nginx** – Web Server
- **MySQL 8.0** – Database Server
- **phpMyAdmin** – Web-based MySQL Administration Tool

Persistent storage is achieved using Docker volumes, while environment variables are used to configure database credentials. The depends on option ensures that dependent services start in the correct order.

Docker Compose Configuration (docker-compose.yml)

```
version: '3.8'

services:
  web:
    image: nginx
    container_name: webserver
    ports:
      - "8090:80"

  database:
    image: mysql:8.0
    container_name: mysqlserver
```

```
    environment:
      MYSQL_ROOT_PASSWORD: root123
      MYSQL_DATABASE: studentdb
      MYSQL_USER: student
      MYSQL_PASSWORD: student123
    ports:
      - "3307:3306"
    volumes:
      - mysql-data:/var/lib/mysql

  phpmyadmin:
    image: phpmyadmin/phpmyadmin
    container_name: phpmyadmin
    environment:
      PMA_HOST: database
      PMA_USER: student
      PMA_PASSWORD: student123
    ports:
      - "8081:80"
    depends_on:
      - database

volumes:
  mysql-data:
```

Procedure

1. Open Docker Desktop and ensure Docker Engine is running.
2. Open Command Prompt or PowerShell.
3. Create a new project directory.

mkdir compose-demo

4. Navigate into the project directory.

cd compose-demo

5. Create a file named **docker-compose.yml**.
6. Copy the above Docker Compose configuration into the file and save it.
7. Verify that the file exists.

dir

8. Deploy the application.

docker compose up -d

9. Verify the running containers.

docker compose ps

10. Open **http://localhost:8090** to verify the Nginx web server.
11. Open **http://localhost:8081** and log in to phpMyAdmin using the MySQL credentials.
12. Validate the compose configuration.

docker compose config

13. Display container logs.

docker compose logs

14. List Docker images used in the project.

docker compose images

15. Stop all services.

docker compose stop

16. Start the services again.

docker compose start

17. Restart all services.

docker compose restart

18. Remove containers and network.

docker compose down

19. Remove containers, network, and volumes.

docker compose down -v

Docker Compose Commands Used

Command	Description
docker compose up -d	Starts all services in detached mode.
docker compose ps	Displays the status of running containers.

Command	Description
docker compose logs	Displays logs of all services.
docker compose images	Lists Docker images used in the project.
docker compose config	Validates and displays the compose configuration.
docker compose stop	Stops all running containers.
docker compose start	Starts previously stopped containers.
docker compose restart	Restarts all services.
docker compose down	Stops and removes containers and the default network.
docker compose down -v	Removes containers, networks, and volumes.

Observations

- Docker Compose successfully created a dedicated network for all services.
- Nginx was accessible through **http://localhost:8090**.
- MySQL database container started successfully with the configured environment variables.
- phpMyAdmin successfully connected to the MySQL database.
- Docker volume **mysql-data** preserved the database data.
- All containers were managed using Docker Compose commands.

Screenshots:

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\L15> docker --version
Docker version 29.6.2, build dfc4efb
PS C:\Users\L15> docker compose version
Docker Compose version v5.3.1
PS C:\Users\L15> mkdir compose-demo


    Directory: C:\Users\L15


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        31-07-2026     23:22                compose-demo


PS C:\Users\L15> cd compose-demo
PS C:\Users\L15\compose-demo> dir
PS C:\Users\L15\compose-demo> notepad docker-compose.yml
PS C:\Users\L15\compose-demo> dir


    Directory: C:\Users\L15\compose-demo

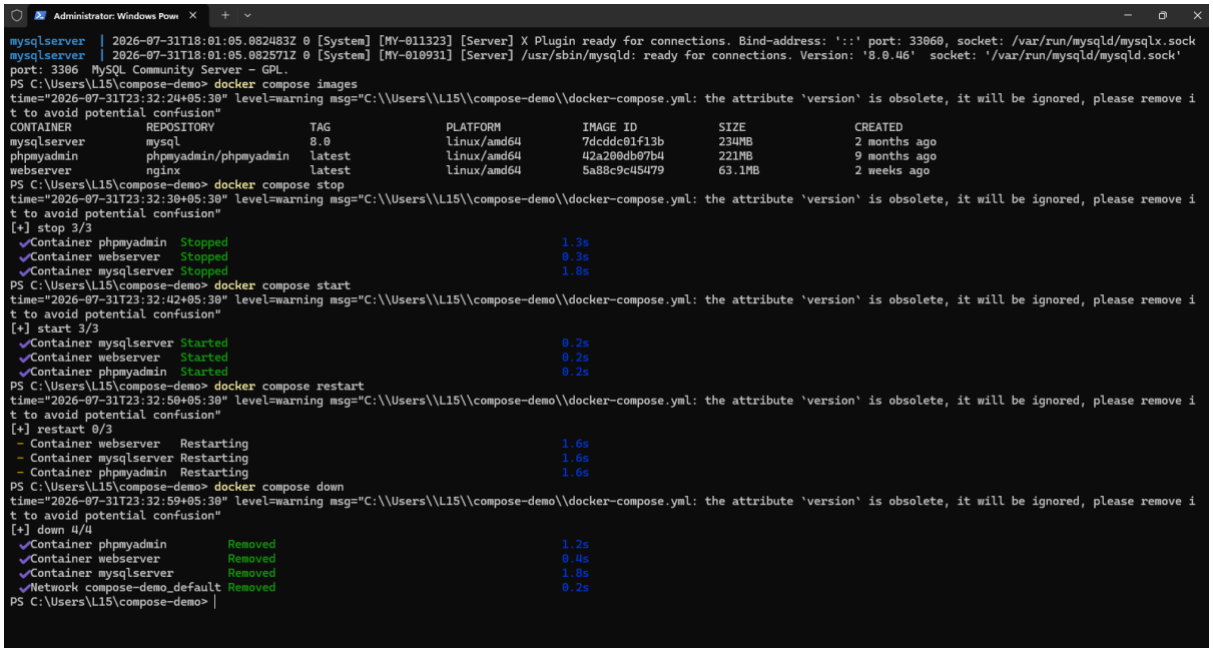

Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----        31-07-2026     23:28            692 docker-compose.yml
```

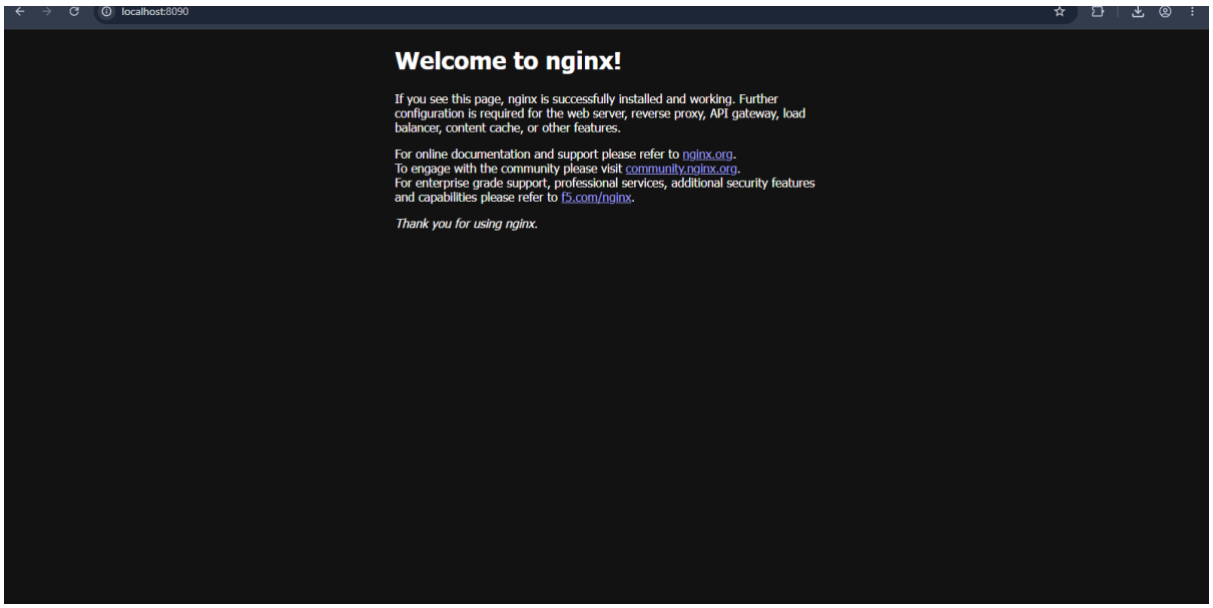
```
PS C:\Users\L15\compose-demo> docker compose up -d
time="2026-07-31T23:29:25+05:30" level=warning msg="C:\\Users\\L15\\compose-demo\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove i
t to avoid potential confusion"
[+] up 51/51
 ✔Image nginx                  Pulled                                  35.8s
 ✔Image mysql:8.0              Pulled                                  82.8s
 ✔Image phpmyadmin/phpmyadmin  Pulled                                  62.1s
 ✔Network compose-demo_default Created                                  0.0s
 ✔Volume compose-demo_mysql-data Created                                0.0s
 ✔Container webserver          Started                                  0.9s
 ✔Container mysqlserver        Started                                  0.9s
 ✔Container phpmyadmin         Started                                  0.5s
PS C:\Users\L15\compose-demo> docker compose ps
time="2026-07-31T23:31:02+05:30" level=warning msg="C:\\Users\\L15\\compose-demo\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove i
t to avoid potential confusion"
NAME          IMAGE                  COMMAND                  SERVICE      CREATED         STATUS         PORTS
mysqlserver   mysql:8.0              "docker-entrypoint.s…"   database     12 seconds ago  Up 11 seconds  0.0.0.0:3307->3306/tcp, [::]:3307->3306/tcp
phpmyadmin    phpmyadmin/phpmyadmin  "/docker-entrypoint.…"   phpmyadmin   11 seconds ago  Up 11 seconds  0.0.0.0:8081->80/tcp, [::]:8081->80/tcp
webserver     nginx                  "/docker-entrypoint.…"   web          12 seconds ago  Up 11 seconds  0.0.0.0:8090->80/tcp, [::]:8090->80/tcp
PS C:\Users\L15\compose-demo> docker compose config
time="2026-07-31T23:32:07+05:30" level=warning msg="C:\\Users\\L15\\compose-demo\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove i
t to avoid potential confusion"
name: compose-demo
services:
  database:
    container_name: mysqlserver
    environment:
      MYSQL_DATABASE: studentdb
      MYSQL_PASSWORD: student123
      MYSQL_ROOT_PASSWORD: root123
      MYSQL_USER: student
    image: mysql:8.0
    networks:
      default: null
    ports:
      - mode: ingress
        target: 3306
        published: "3307"
        protocol: tcp
    volumes:
```

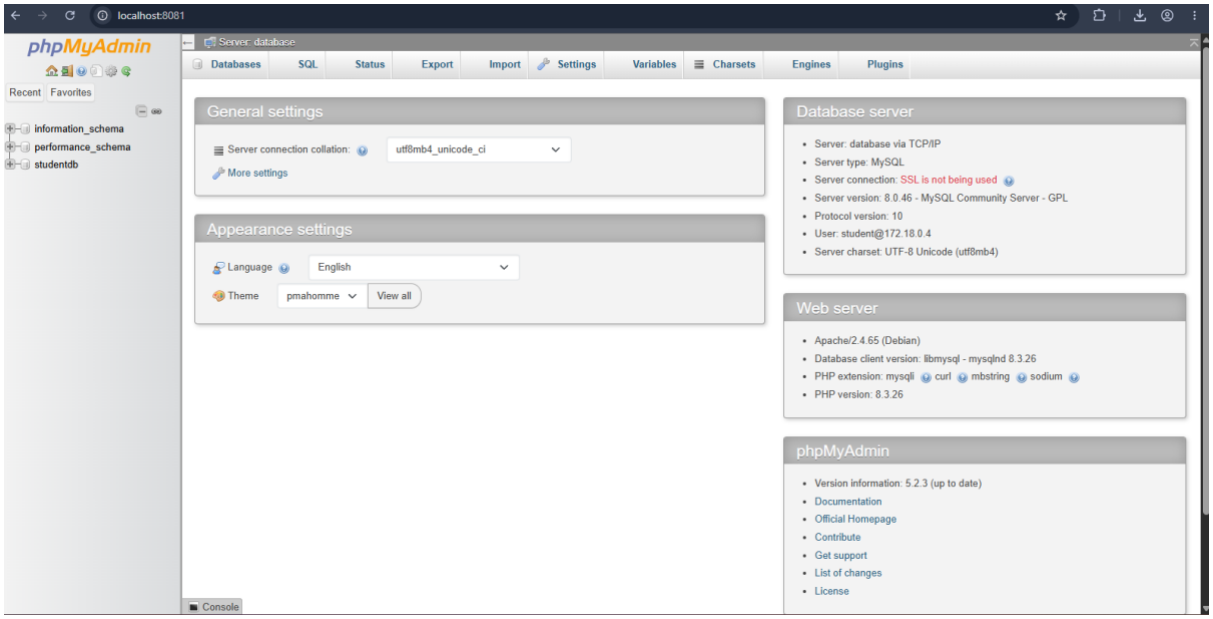
```
  mysql-data:
    name: compose-demo_mysql-data
PS C:\Users\L15\compose-demo> docker compose logs
time="2026-07-31T23:32:16+05:30" level=warning msg="C:\\Users\\L15\\compose-demo\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove i
t to avoid potential confusion"
mysqlserver  | 2026-07-31 18:00:51+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 8.0.46-1.el9 started.
mysqlserver  | 2026-07-31 18:00:51+00:00 [Note] [Entrypoint]: Switching to dedicated user 'mysql'
mysqlserver  | 2026-07-31 18:00:51+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 8.0.46-1.el9 started.
mysqlserver  | 2026-07-31 18:00:51+00:00 [Note] [Entrypoint]: Initializing database files
phpmyadmin   | AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 172.18.0.4. Set the 'ServerName' directive globally to suppress
this message
mysqlserver  | 2026-07-31T18:00:51.710866Z 0 [Warning] [MY-011068] [Server] The syntax '--skip-host-cache' is deprecated and will be removed in a future release. Please use S
ET GLOBAL host_cache_size=0 instead.
mysqlserver  | 2026-07-31T18:00:51.710958Z 0 [System] [MY-013169] [Server] /usr/sbin/mysqld (mysqld 8.0.46) initializing of server in progress as process 80
mysqlserver  | 2026-07-31T18:00:51.717918Z 1 [System] [MY-013576] [InnoDB] InnoDB initialization has started.
mysqlserver  | 2026-07-31T18:00:52.648476Z 1 [System] [MY-013577] [InnoDB] InnoDB initialization has ended.
mysqlserver  | 2026-07-31T18:00:54.420651Z 6 [Warning] [MY-010453] [Server] root@localhost is created with an empty password ! Please consider switching off the --initialize-
insecure option.
mysqlserver  | 2026-07-31 18:00:59+00:00 [Note] [Entrypoint]: Database files initialized
mysqlserver  | 2026-07-31 18:00:59+00:00 [Note] [Entrypoint]: Starting temporary server
mysqlserver  | 2026-07-31T18:00:59.530038Z 0 [Warning] [MY-011068] [Server] The syntax '--skip-host-cache' is deprecated and will be removed in a future release. Please use S
ET GLOBAL host_cache_size=0 instead.
mysqlserver  | 2026-07-31T18:00:59.531396Z 0 [System] [MY-010116] [Server] /usr/sbin/mysqld (mysqld 8.0.46) starting as process 124
mysqlserver  | 2026-07-31T18:00:59.543947Z 1 [System] [MY-013576] [InnoDB] InnoDB initialization has started.
mysqlserver  | 2026-07-31T18:00:59.823671Z 1 [System] [MY-013577] [InnoDB] InnoDB initialization has ended.
mysqlserver  | 2026-07-31T18:01:00.071040Z 0 [Warning] [MY-010068] [Server] CA certificate ca.pem is self signed.
mysqlserver  | 2026-07-31T18:01:00.071074Z 0 [System] [MY-013602] [Server] Channel mysql_main configured to support TLS. Encrypted connections are now supported for this chan
nel.
mysqlserver  | 2026-07-31T18:01:00.075092Z 0 [Warning] [MY-011810] [Server] Insecure configuration for --pid-file: Location '/var/run/mysqld' in the path is accessible to all
 OS users. Consider choosing a different directory.
phpmyadmin   | AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 172.18.0.4. Set the 'ServerName' directive globally to suppress
this message
mysqlserver  | 2026-07-31T18:01:00.089040Z 0 [System] [MY-011323] [Server] X Plugin ready for connections. Socket: /var/run/mysqld/mysqlx.sock
phpmyadmin   | [Fri Jul 31 18:00:51.700946 2026] [mpm_prefork:notice] [pid 1:tid 1] AH00163: Apache/2.4.65 (Debian) PHP/8.3.26 configured -- resuming normal operations
mysqlserver  | 2026-07-31T18:01:00.089127Z 0 [System] [MY-010931] [Server] /usr/sbin/mysqld: ready for connections. Version: '8.0.46'  socket: '/var/run/mysqld/mysqld.sock'
port: 0  MySQL Community Server - GPL.
mysqlserver  | 2026-07-31 18:01:00+00:00 [Note] [Entrypoint]: Temporary server started.
mysqlserver  | '/var/lib/mysql/mysql.sock' -> '/var/run/mysqld/mysqld.sock'
mysqlserver  | Warning: Unable to load '/usr/share/zoneinfo/iso3166.tab' as time zone. Skipping it.
mysqlserver  | Warning: Unable to load '/usr/share/zoneinfo/leap-seconds.list' as time zone. Skipping it.
mysqlserver  | Warning: Unable to load '/usr/share/zoneinfo/leapseconds' as time zone. Skipping it.
mysqlserver  | Warning: Unable to load '/usr/share/zoneinfo/tzdata.zi' as time zone. Skipping it.
```




```
Administrator: Windows Powe
.3" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/150.0.0.0 Safari/537.36"
phpmyadmin   | 172.18.0.1 - - [31/Jul/2026:18:01:41 +0000] "GET /themes/pmahomme/img/s_lang.png HTTP/1.1" 200 943 "http://localhost:8081/themes/pmahomme/css/theme.css?v=5.2.3
" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/150.0.0.0 Safari/537.36"
phpmyadmin   | 172.18.0.1 - - [31/Jul/2026:18:01:41 +0000] "GET /themes/pmahomme/img/s_theme.png HTTP/1.1" 200 1056 "http://localhost:8081/themes/pmahomme/css/theme.css?v=5.2
.3" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/150.0.0.0 Safari/537.36"
phpmyadmin   | 172.18.0.1 - - [31/Jul/2026:18:01:41 +0000] "GET /themes/pmahomme/img/s_notice.png HTTP/1.1" 200 851 "http://localhost:8081/themes/pmahomme/css/theme.css?v=5.2
.3" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/150.0.0.0 Safari/537.36"
phpmyadmin   | 172.18.0.1 - - [31/Jul/2026:18:01:41 +0000] "GET /themes/pmahomme/img/window-new.png HTTP/1.1" 200 752 "http://localhost:8081/themes/pmahomme/css/theme.css?v=5
.2.3" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/150.0.0.0 Safari/537.36"
phpmyadmin   | 172.18.0.1 - - [31/Jul/2026:18:01:41 +0000] "POST /index.php?route=/config/get HTTP/1.1" 200 2012 "http://localhost:8081/" "Mozilla/5.0 (Windows NT 10.0; Win64
; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/150.0.0.0 Safari/537.36"
phpmyadmin   | 172.18.0.1 - - [31/Jul/2026:18:01:41 +0000] "GET /favicon.ico HTTP/1.1" 200 22788 "http://localhost:8081/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebK
it/537.36 (KHTML, like Gecko) Chrome/150.0.0.0 Safari/537.36"
phpmyadmin   | 172.18.0.1 - - [31/Jul/2026:18:01:41 +0000] "POST /index.php?route=/navigation&ajax_request=1 HTTP/1.1" 200 2631 "http://localhost:8081/" "Mozilla/5.0 (Windows
 NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/150.0.0.0 Safari/537.36"
phpmyadmin   | 172.18.0.1 - - [31/Jul/2026:18:01:41 +0000] "POST /index.php?route=/config/get HTTP/1.1" 200 2104 "http://localhost:8081/" "Mozilla/5.0 (Windows NT 10.0; Win64
; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/150.0.0.0 Safari/537.36"
phpmyadmin   | 172.18.0.1 - - [31/Jul/2026:18:01:41 +0000] "POST /index.php?route=/version-check HTTP/1.1" 200 684 "http://localhost:8081/" "Mozilla/5.0 (Windows NT 10.0; Win
64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/150.0.0.0 Safari/537.36"
phpmyadmin   | 172.18.0.1 - - [31/Jul/2026:18:01:41 +0000] "POST /index.php?route=/config/set HTTP/1.1" 200 2004 "http://localhost:8081/" "Mozilla/5.0 (Windows NT 10.0; Win64
; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/150.0.0.0 Safari/537.36"
phpmyadmin   | 127.0.0.1 - - [31/Jul/2026:18:01:47 +0000] "OPTIONS * HTTP/1.0" 200 126 "-" "Apache/2.4.65 (Debian) PHP/8.3.26 (internal dummy connection)"
phpmyadmin   | 127.0.0.1 - - [31/Jul/2026:18:01:48 +0000] "OPTIONS * HTTP/1.0" 200 126 "-" "Apache/2.4.65 (Debian) PHP/8.3.26 (internal dummy connection)"
phpmyadmin   | 172.18.0.1 - - [31/Jul/2026:18:01:52 +0000] "POST /index.php?route=/config/set HTTP/1.1" 200 2005 "http://localhost:8081/" "Mozilla/5.0 (Windows NT 10.0; Win64
; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/150.0.0.0 Safari/537.36"
phpmyadmin   | 172.18.0.1 - - [31/Jul/2026:18:01:54 +0000] "POST /index.php?route=/config/set HTTP/1.1" 200 2004 "http://localhost:8081/" "Mozilla/5.0 (Windows NT 10.0; Win64
; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/150.0.0.0 Safari/537.36"
mysqlserver  | 2026-07-31T18:01:03.404533Z 0 [System] [MY-010910] [Server] /usr/sbin/mysqld: Shutdown complete (mysqld 8.0.46)  MySQL Community Server - GPL.
mysqlserver  | 2026-07-31 18:01:04+00:00 [Note] [Entrypoint]: Temporary server stopped
mysqlserver  |
mysqlserver  | 2026-07-31 18:01:04+00:00 [Note] [Entrypoint]: MySQL init process done. Ready for start up.
mysqlserver  |
mysqlserver  | 2026-07-31T18:01:04.595751Z 0 [Warning] [MY-011068] [Server] The syntax '--skip-host-cache' is deprecated and will be removed in a future release. Please use S
ET GLOBAL host_cache_size=0 instead.
mysqlserver  | 2026-07-31T18:01:04.596611Z 0 [System] [MY-010116] [Server] /usr/sbin/mysqld (mysqld 8.0.46) starting as process 1
mysqlserver  | 2026-07-31T18:01:04.601275Z 1 [System] [MY-013576] [InnoDB] InnoDB initialization has started.
mysqlserver  | 2026-07-31T18:01:04.873892Z 1 [System] [MY-013577] [InnoDB] InnoDB initialization has ended.
mysqlserver  | 2026-07-31T18:01:05.071290Z 0 [Warning] [MY-010068] [Server] CA certificate ca.pem is self signed.
mysqlserver  | 2026-07-31T18:01:05.071326Z 0 [System] [MY-013602] [Server] Channel mysql_main configured to support TLS. Encrypted connections are now supported for this chan
nel.
mysqlserver  | 2026-07-31T18:01:05.075539Z 0 [Warning] [MY-011810] [Server] Insecure configuration for --pid-file: Location '/var/run/mysqld' in the path is accessible to all
 OS users. Consider choosing a different directory.
```


Administrator: Windows Powe
mysqlserver | 2026-07-31T18:01:05.082483Z 0 [System] [MY-011323] [Server] X Plugin ready for connections. Bind-address: '::' port: 33060, socket: /var/run/mysqld/mysqlx.sock
mysqlserver | 2026-07-31T18:01:05.082571Z 0 [System] [MY-010931] [Server] /usr/sbin/mysqld: ready for connections. Version: '8.0.46' socket: '/var/run/mysqld/mysqld.sock'
port: 3306 MySQL Community Server - GPL.
PS C:\Users\L15\compose-demo> docker compose images
time="2026-07-31T23:32:24+05:30" level=warning msg="C:\\Users\\L15\\compose-demo\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"
CONTAINER REPOSITORY TAG PLATFORM IMAGE ID SIZE CREATED
mysqlserver mysql 8.0 linux/amd64 7dcddc01f13b 234MB 2 months ago
phpmyadmin phpmyadmin/phpmyadmin latest linux/amd64 42a200db07b4 221MB 9 months ago
webserver nginx latest linux/amd64 5a88c9c45479 63.1MB 2 weeks ago
PS C:\Users\L15\compose-demo> docker compose stop
time="2026-07-31T23:32:30+05:30" level=warning msg="C:\\Users\\L15\\compose-demo\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"
[+] stop 3/3
Container phpmyadmin Stopped 1.3s
Container webserver Stopped 0.3s
Container mysqlserver Stopped 1.8s
PS C:\Users\L15\compose-demo> docker compose start
time="2026-07-31T23:32:42+05:30" level=warning msg="C:\\Users\\L15\\compose-demo\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"
[+] start 3/3
Container mysqlserver Started 0.2s
Container webserver Started 0.2s
Container phpmyadmin Started 0.2s
PS C:\Users\L15\compose-demo> docker compose restart
time="2026-07-31T23:32:50+05:30" level=warning msg="C:\\Users\\L15\\compose-demo\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"
[+] restart 0/3
Container webserver Restarting 1.6s
Container mysqlserver Restarting 1.6s
Container phpmyadmin Restarting 1.6s
PS C:\Users\L15\compose-demo> docker compose down
time="2026-07-31T23:32:59+05:30" level=warning msg="C:\\Users\\L15\\compose-demo\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"
[+] down 4/4
Container phpmyadmin Removed 1.2s
Container webserver Removed 0.4s
Container mysqlserver Removed 1.8s
Network compose-demo_default Removed 0.2s
PS C:\Users\L15\compose-demo>


localhost:8090
Welcome to nginx!
If you see this page, nginx is successfully installed and working. Further configuration is required for the web server, reverse proxy, API gateway, load balancer, content cache, or other features.
For online documentation and support please refer to nginx.org.
To engage with the community please visit community.nginx.org.
For enterprise grade support, professional services, additional security features and capabilities please refer to f5.com/nginx.
Thank you for using nginx.



Result

The multi-service application consisting of Nginx, MySQL, and phpMyAdmin was successfully deployed using Docker Compose. All services started correctly, communicated over the Docker Compose network, and were managed using Docker Compose commands.

Conclusion

Docker Compose simplifies the deployment and management of multi-container applications by defining all services in a single configuration file. It enables automatic networking, persistent storage using Docker volumes, simplified container orchestration, and easy deployment using a single command. This experiment demonstrated efficient deployment and management of multiple interdependent services using Docker Compose.